

FARZANA ISLAM ADIBA

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Important Links: Google Scholar—ResearchGate—GitHub

EDUCATION

Ph.D. in Computer Science

Department of Computer and Information Sciences, University of Delaware

Fall 2024 – Present

M.S. in AI in Healthcare Initiatives

University of Florida

Spring 2024

M.Sc in Computer Science and Engineering (CSE)

Jahangirnagar University, Dhaka, Bangladesh

2019 – 2022

B.Sc in Computer Science and Engineering (CSE)

Jahangirnagar University, Dhaka, Bangladesh

2015 – 2019

RESEARCH INTEREST

- Bias and Fairness in Medical LLMs
- AI for Health Equity
- Clinical Decision Support
- Health Informatics

RESEARCH EXPERIENCE

University of Delaware; Graduate Research Assistant

September 2024 – Present

- Working on Med LLMs' Bias Evals, and Mitigation Approaches

University of Florida; Graduate Research Assistant

Jan 2023 – May 2024

- Developed deep learning pipelines to identify pediatric fall events and cannabis usage (Data: PRECEDE, UF)
- Applied pre-trained language models for sentence similarity to detect suicidal keywords (Data: MIMIC-III/IV)
- Extracted postpartum depression keywords using few-shot LLMs (LLaMA2, BERT).

Jahangirnagar University; Graduate Research Assistant

Sept 2019 – February 2022

- Implemented deep learning models for multivariate time-series prediction using Pediatric ICU data (Data: PICU, National Heart Foundation, Bangladesh).
- Developed ML/DL classification models to predict neonatal ICU length of stay based on drug and dosage analysis (Data: MIMIC-III).

TEACHING EXPERIENCE

Graduate Teaching Assistant

Course: CISC220 (Data Structures)

University of Delaware

Fall 2024 - Spring 2025

Graduate Teaching Assistant

Course: PHA6805 (Applied Data Analysis and Interpretation)

University of Florida, Gainesville, FL

Fall 2023

Lecturer, Dept of Computer Science and Engineering (Study leave)

University of Information Technology & Sciences (UITS), Dhaka, Bangladesh

April 2022 – Present

Courses I Taught

- Numerical Method Analysis
- Computer Networks Lab
- Linux Programming Lab
- Database Management System Lab
- Structured Programming Language Lab (Java)

PUBLICATIONS

Journal Papers:

- Ruba Sajdeya, Mamoun T Mardini, Patrick J Tighe, Ronald L Ison, Chen Bai, Sebastian Jugl, Gao Hanzhi, Kimia Zandbiglari, **Farzana Islam Adiba**, Almut G Winterstein, Thomas A Pearson, Robert L Cook, Masoud Rouhizadeh. “Developing and validating a natural language processing algorithm to extract preoperative cannabis use status documentation from unstructured narrative clinical notes,” Journal of the American Medical Informatics Association, Volume 30, Issue 8, August 2023, Pages 1418–1428, <https://doi.org/10.1093/jamia/ocad080>
- **Farzana Islam Adiba**, Tahmina Islam, M Shamim Kaiser, Mufti Mahmud, Muhammad Arifur Rahman. “Effect of corpora on classification of fake news using Naive Bayes classifier.” Int. J. Autom. AI Mach. Learn. (IJAAIML) 1 (1) (2020). <https://researchlakejournals.com/index.php/AAIML/article/view/45/32>

Conference Papers:

- **Farzana Islam Adiba**, Mohammad Zahidur Rahman. “Machine Learning Models to Analyze the Effect of Drugs on Neonatal-ICU Length of Stay.” Applied Intelligence and Informatics(AII 2022). Communications in Computer and Information Science, vol 1724. Springer, Cham. https://doi.org/10.1007/978-3-031-24801-6_14
- **Farzana Islam Adiba**, Sharmin Nahar Sharwardy, Mohammad Zahidur Rahman, “Multivariate Time Series Prediction of Pediatric ICU data using Deep Learning,” 2021 International Conference on Innovative Trends in Information Technology (ICITIIT), Kottayam, India, 2021, pp. 1-6, doi:10.1109/ICITIIT51526.2021.9399593.

Poster Presentation:

- **Farzana Islam Adiba**, Masoud Rouhizadeh. “Large Language Models for Identifying Social Determinants of Health in Postpartum Depression” 37th Annual Research Showcase, College of Pharmacy, University of Florida.
- **Farzana Islam Adiba**, Elise Omaki, Kimia Zandbiglari, Wendy C. Shields, Andrea C Gielen, Masoud Rouhizadeh. “Deep Learning and Natural Language Processing Methods for Characterizing Pediatric Falls Events.” AMIA 2023 Annual Symposium, New Orleans, LA.
- **Farzana Islam Adiba**, Mohammad Zahidur Rahman. “Design and Development of Spectrometer for Chromatography.” 6th International Conference on Networking, Systems, and Security (6th NSysS 2019), BUET, Dhaka, Bangladesh.

Preprint:

- **Farzana Islam Adiba**, Varsha Danduri, Fahmida Liza Piya, Ali Abbasi, Mehak Gupta, and Rahmatollah Beheshti. “A Multimodal Data Processing Pipeline for MIMIC-IV Dataset.” arXiv preprint arXiv:2601.11606 (2026).
- **Farzana Islam Adiba**, and Rahmatollah Beheshti. “Tell Me You’re Biased Without Telling Me You’re Biased—Toward Revealing Implicit Biases in Medical LLMs.” arXiv preprint arXiv:2507.21176 (2025).
- **Farzana Islam Adiba**, Yifan Zhang, and Rahmatollah Beheshti. (2025, June 30). Bias and Fairness in Medical LLMs: An Extensive Scoping Review. doi:10.31219/osf.io/fqejh_v1.
- *Elise Omaki, ***Farzana Islam Adiba**, Muhammad Bilal, Shobhan Kumar, Kimia Zandbiglari, Masoud Rouhizadeh. “A Comparative Study of Machine Learning and Deep Learning Techniques for Classifying Pediatric Fall Injuries Using Natural Language Processing on NEISS Narratives.”

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C/C++, SQL
- **ML tools:** TensorFlow, PyTorch, SciKit Learn, Pandas, NumPy, NLTK, spaCy, HuggingFace Transformers
- **Software and Tools:** Anaconda, Jupyter Notebook, Google Colab, Visual Studio, Arduino IDE, Android Studio, R Studio, git Bash, L^AT_EX
- **Virtual Environment and Repositories:** Ubuntu Virtual Machine, Docker, GitHub

AWARDS & ACHIEVEMENTS

- Pharmaceutical Outcomes and Policy (POP) Programmer Award 2023
Issued by: Department of Pharmaceutical Outcomes and Policy (POP), University of Florida
- National Science and Technology Fellowship (NST Fellowship 2020 - 2021) for M.Sc Thesis, Jan 2021
Awarded by: Ministry of Science and Technology, Government of the People’s Republic of Bangladesh
- Bangladesh Government Scholarship for overall academic results during undergraduate (2015-2019).
- Jahangirnagar University Supplementary Scholarship on academic results during undergraduate (2015-2019).
- Jahangirnagar University Scholarship for achieving the **1st position** in the undergraduate admission test 2014-2015.

SERVICES

- Program Committee, AAAI 2027 Artificial Intelligence for Social Impact Track [2027]
- Reviewer, AHLI Conference on Health, Inference, and Learning (CHIL), Berkeley, California, [2025 - Present]
- Reviewer, Machine Learning for Healthcare (MLHC), [2024 - Present]
- Reviewer, Workshop on Deploying AI in Healthcare (DAIH) at Conference on Language Modeling (COLM 2026)
- Reviewer, Machine Learning for Health Symposium 2025 (ML4H 2025), San Diego, CA
- Reviewer, AMIA Informatics Conference [2023 - Present]
- Program Committee, Workshop on Bangla Language Processing, Empirical Methods in Natural Language Processing (ACL) [2023 - Present]
- Co-chair, Journal Club, Department of Pharmaceutical Outcomes and Policy, Fall 2023.